

SLOPE STABILITY OF TANGENT BUNDLES OF SMOOTH TORIC FANO VARIETIES

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ABSTRACT. We classify the anticanonical slope stability of tangent bundles for all 8,630 smooth toric Fano varieties in dimensions three through six, by exact evaluation of Klyachko’s criterion, and determine polystability (equivalently, the existence of a Hermitian–Einstein metric with respect to an anticanonical Kähler form) in every strictly semistable case. Every Kähler–Einstein variety in the census has polystable tangent bundle, whereas the converse fails widely: 102 of the 109 five-folds with stable tangent bundle are not Kähler–Einstein.

The census singles out one construction at high Picard rank, which we introduce in general: root-twisted toric dP_3 -fibrations over products of projective lines, parametrized by roots of A_2 . Every nonempty multiset of nonzero root twists with vanishing sum produces a stable tangent bundle, in every dimension. Among these zero-sum twists, the resulting variety is Kähler–Einstein if and only if the multiset is invariant under negation or under the order-three rotation of the root hexagon. Thus stable toric Fanos of Picard rank $n + 2$ exist for every $n \geq 4$; vanishing twist sum does not force the Kähler–Einstein property, while a separate unbalanced family shows that a vanishing sum is not necessary for stability.

The proof reduces the slope inequalities for the root-twist family to integrals over the A_2 moment hexagon. Positive layer decompositions and sharp one-dimensional convolution estimates establish stability, while a strict ordering of the hexagon’s first moments at the extreme exponents yields the Kähler–Einstein classification.

1. INTRODUCTION

Let X be a Fano manifold of dimension n , polarized by $-K_X$. Two conditions of an Einstein type compete on X : the existence of a Kähler–Einstein metric on the manifold, and the existence of a Hermitian–Einstein metric on the tangent bundle with respect to a Kähler form in $c_1(X)$. By the Kobayashi–Hitchin correspondence of Donaldson and Uhlenbeck–Yau [Don85, UY86], the second condition is equivalent to the polystability of T_X in the sense of Mumford–Takemoto slopes; and if X is Kähler–Einstein, the Kähler–Einstein metric is itself Hermitian–Einstein on T_X , so Kähler–Einstein implies polystable [Kob87, Lüb83]. The converse fails: Fahlaoui [Fah89] showed that the blow-up of $\mathbb{P}^2$ in two points has stable tangent bundle, although by Matsushima’s theorem [Mat57] it admits no Kähler–Einstein metric. The relationship between the two conditions beyond this single implication has remained largely unexplored, for a simple reason: outside of homogeneous and low-dimensional examples, deciding the stability of a tangent bundle is hard.

Toric varieties are the exception. Klyachko’s classification of equivariant vector bundles [Kly90], in the form developed by Perling and Kool for reflexive sheaves [Per04, Koo11], reduces slope inequalities for equivariant sheaves on a smooth toric variety to finitely many

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exact comparisons of lattice quantities. Hering, Nill and Süß [HNS22] distilled this into a clean combinatorial criterion for tangent bundles (Proposition 2.1 below) and used it to classify stability of T_X over the full ample cone for all smooth toric surfaces and all smooth toric varieties of Picard rank 2. On the Kähler–Einstein side, the toric case is equally decidable: by Mabuchi and Wang–Zhu [Mab87, WZ04], a smooth toric Fano is Kähler–Einstein if and only if the barycenter of its moment polytope vanishes.

Complete classification lists are available in all dimensions of our census: 5 surfaces, 18 three-folds [WW82, Bat99], 124 four-folds [Bat99, Sat00], and, by Øbro’s algorithm [Øbr07], 866 five-folds and 7622 six-folds. Thus both sides of the comparison are, in principle, finite computations across the entire landscape. In practice the literature covers: dimension 2 [Fah89, HNS22]; dimension 3, where Steffens [Ste96] classified $(-K)$ -stability of T_X for *all* smooth Fano three-folds; and dimension 4 for Picard rank $\rho(X) \leq 2$ (Biswas–Dey–Genç–Poddar [BDGP21]) and $\rho \leq 3$ (Dasgupta–Dey–Khan [DDK20]). The four-folds of Picard rank ≥ 4 (86 of Batyrev’s 124) and everything in dimensions five and six were open.

This paper closes the gap through dimension six and carries out the stability–Kähler–Einstein comparison across the resulting census. Our main results are the following.

Theorem 1.1 (Classification; Theorem 4.1). *Of the smooth toric Fano varieties of dimension $n = 3, 4, 5, 6$, exactly 4, 26, 109, 636 have $(-K)$ -stable tangent bundle; exactly 7, 24, 82, 366 are strictly semistable; and exactly 7, 74, 675, 6620 are unstable.*

Theorem 1.2 (Hermitian–Einstein classification; Theorem 5.4). *Exactly 8 of the 18 smooth toric Fano three-folds, 40 of the 124 four-folds, 161 of the 866 five-folds, and 895 of the 7622 six-folds admit a Hermitian–Einstein metric on their tangent bundle with respect to a Kähler form representing $c_1(X)$.*

The passage from the classification to the census requires deciding, for each strictly semistable T_X , whether it is polystable. For tangent bundles of toric varieties this is again purely combinatorial: polystability of an equivariant reflexive sheaf can always be exhibited by an equivariant splitting (Lemma 5.1, proved for arbitrary polystable equivariant reflexive sheaves and perhaps of independent use), and the two-step structure of the Klyachko filtrations then implies that equivariant direct-sum decompositions of T_X correspond exactly to partitions of the ray generators into groups spanning complementary subspaces (Proposition 5.2); polystability becomes a finite search over such partitions. More canonically, the connected components of the ray matroid give the finest splitting and force a product decomposition of the variety. Consequently a strictly semistable tangent bundle is polystable exactly when the variety is a nontrivial product of smooth toric Fanos with stable tangent bundles (Corollary 5.3). The refinement is geometrically meaningful. For instance, the blow-up of $\mathbb{P}^3$ along a line is strictly semistable but not polystable (the three-dimensional analogue of the Hirzebruch surface $\mathbb{F}_1$), and thus carries no Hermitian–Einstein metric despite being semistable; while $d\mathbb{P}_2 \times \mathbb{P}^1$ is polystable, giving a Hermitian–Einstein tangent bundle on a manifold with no Kähler–Einstein metric.

The data exhibit three phenomena.

First, instability is the dominant condition, and increasingly so with dimension: the unstable fraction grows from 7/18 in dimension three to 74/124 in dimension four to 675/866 in dimension five. The Hermitian–Einstein locus thins correspondingly.

Second, Kähler–Einstein is a much rarer condition than a Hermitian–Einstein tangent bundle, and the gap widens: in dimension five, 109 varieties have stable T_X but only 7 of them are Kähler–Einstein. Fahlaoui’s surface example is not an exception; it is the first instance of a widespread pattern in the census. In the other direction our data confirm, in

every dimension computed, that all Kähler–Einstein varieties have polystable tangent bundle, as the Kobayashi–Hitchin correspondence requires; this includes its combinatorial form, that a vanishing barycenter forces the subspace concentration inequalities of [HNS22, HL14].

Third, at high Picard rank stability collapses to a single, structured example. Among the 42 five-folds with $\rho \geq 7$ exactly one is stable. It is Kähler–Einstein, and it has a transparent geometric description: a toric dP_3 -fibration over $(\mathbb{P}^1)^3$ in which the three base directions are twisted by the three roots $\alpha, \beta, -(\alpha + \beta)$ of A_2 . These three twists form an orbit of the order-three rotation of the root hexagon, a symmetry that forces the barycenter to vanish (hence the Kähler–Einstein property), while the twist destroys the direct-sum decomposition that makes the untwisted product $\mathrm{dP}_3 \times (\mathbb{P}^1)^3$ merely polystable: it turns every slope equality into a strict inequality. We define this root-twist construction for every number of base factors (Section 7). Exact computation first verified zero-sum stability through dimension eight; Theorem 7.2 proves it in every dimension. Theorem 7.3 further proves that such a variety is Kähler–Einstein exactly when its twist multiset is symmetric under negation or the order-three rotation. A vanishing sum is not necessary for stability: we prove that the unbalanced twists $((\alpha, -\alpha)^{\times r}, \alpha, \beta)$ are stable and non-Kähler–Einstein for every $r \geq 0$. Together with the two symmetric families, whose twists form a repeated $\pm\alpha$ pair or a repeated rotation orbit, this gives three particularly explicit infinite families. The proofs rest on an exact criterion (Proposition 7.6): every slope inequality of the family is an inequality between integrals over the two-dimensional moment hexagon of dP_3 , and stability is equivalent to the negativity of one such integral per twist together with a strict triangle inequality among three line shares, one attached to each root line of the hexagon. In the general zero-sum case these inequalities follow from centered-box convolution estimates.

Theorem 1.3 (Zero-sum root twists; Theorems 7.2 and 7.3). *Let $t_1, \dots, t_k$ be nonzero roots of A_2 with $\sum_i t_i = 0$. Then $T_{X(t_\bullet)}$ is $(-K)$ -stable. Consequently, for every $n \geq 4$ there is a smooth toric Fano n -fold of Picard rank $n + 2$ with stable tangent bundle. Moreover, $X(t_\bullet)$ is Kähler–Einstein if and only if the twist multiset is invariant under negation or under the order-three rotation of the root hexagon.*

Theorem 1.4 (Three infinite families; Theorem 7.11). *For every $r \geq 1$, $m \geq 1$, and $s \geq 0$, the root-twist varieties $X((\alpha, -\alpha)^{\times r})$, of dimension $n = 2r + 2$, and $X((\alpha, \beta, -(\alpha + \beta))^{\times m})$, of dimension $n = 3m + 2$, have $(-K)$ -stable tangent bundle and are Kähler–Einstein. The varieties $X((\alpha, -\alpha)^{\times s}, \alpha, \beta)$, of dimension $n = 2s + 4$, also have stable tangent bundle but are not Kähler–Einstein. All three families are smooth toric Fanos of Picard rank $n + 2$. In particular, stable Kähler–Einstein examples of rank $n + 2$ exist in every even dimension $n \geq 4$ and every dimension $n \equiv 2 \pmod{3}$, $n \geq 5$, while stable non-Kähler–Einstein examples of that rank exist in every even dimension $n \geq 4$.*

Finally, for every unstable variety the computation records the canonical first Harder–Narasimhan subspace. These subspaces have structure: in 7/7, 68/74, 558/675 and 4877/6620 of the unstable cases in dimensions 3, 4, 5, 6, they contain an antipodal pair of rays. Such a pair is the combinatorial signature of a candidate relative tangent direction for an equivariant $\mathbb{P}^1$ -fibration, in line with Steffens’s principle that destabilizing subsheaves of Fano tangent bundles arise from relative tangent sheaves of extremal contractions [Ste96]. Identifying the corresponding contractions is a separate geometric step.

Related and concurrent work. The combinatorial machinery of equivariant reflexive sheaves on toric varieties has developed rapidly alongside this work: Napame and Tipler

study slope stability of reflexive pullbacks along toric fibrations and produce stable perturbations of semistable tangent sheaves by varying the polarization or blowing up [NT24]; Clarke, Napame and Tipler analyse how slope (poly)stability of toric sheaves behaves under toric flips [CNT25]; Tipler studies the Chern classes of stable toric sheaves [Tip25]; and Delloque, Napame, Scarpa and Tipler develop polynomial stability conditions with equivariant positivity on T -varieties [DNST25]. On the Fano side, Kanemitsu [Kan21] determined the stability of the tangent bundles of Pasquier’s two-orbit varieties, disproving the long-standing expectation that a Fano manifold of Picard number one has stable tangent bundle; and concurrently with the present paper, Chen and Lai [CL26] characterize the maximal destabilizing sheaf of the tangent sheaf of a Fano variety through the foliated minimal model program, classifying the $(-K)$ -slope-unstable weak del Pezzo surfaces and exhibiting a singular del Pezzo surface that carries a weak Kähler–Einstein metric yet has slope-unstable tangent sheaf. In the singular setting, therefore, a weak Kähler–Einstein metric need not force slope-semistability of T_X .

Reliability. Because the mathematical content of a classification of this kind is concentrated in the correctness of the computation, we describe the validation in detail (Section 3). In brief: all arithmetic is exact (integer and rational); every variety passes internal consistency identities; and the implementation reproduces, without discrepancy, every published slice of the problem (Fahlaoui’s del Pezzo table; the toric slice of Steffens’s classification of Fano three-folds; the 38 four-folds of Picard rank ≤ 3 classified by [BDGP21, DDK20], matched variety-by-variety via exact $GL(4, \mathbb{Z})$ -equivalence of fans; and the Kähler–Einstein tables of [Nak93, Nak94, NSY23]). All analysis code, the input polytope data, and the per-variety result files are publicly available at <https://github.com/bwuebben/toric-tangent-stability>; every verdict is independently recomputable from the ray data alone.

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2. PRELIMINARIES

2.1. Smooth toric Fano varieties. Let $N \cong \mathbb{Z}^n$ be a lattice, M its dual. A smooth toric Fano n -fold $X = X_\Sigma$ corresponds to a *smooth Fano polytope*: the convex hull, in $N_\mathbb{Q}$, of the primitive ray generators u_ρ of Σ , which is reflexive with every facet a unimodular simplex. The dual reflexive polytope $P = \{m \in M_\mathbb{Q} : \langle m, u_\rho \rangle \geq -1 \ \forall \rho\}$ is the moment polytope of $-K_X$. For a facet $F_\rho = P \cap \{\langle \cdot, u_\rho \rangle = -1\}$ we write $\text{Vol}(F_\rho)$ for its normalized lattice volume, i.e. $(n-1)!$ times its Euclidean volume with respect to the induced lattice; then

$$(1) \quad \deg_{-K}(D_\rho) = (-K_X)^{n-1} \cdot D_\rho = \text{Vol}(F_\rho), \quad \sum_{\rho} \text{Vol}(F_\rho) = (-K_X)^n = n! \text{vol}(P).$$

By Mabuchi and Wang–Zhu [Mab87, WZ04], X admits a Kähler–Einstein metric if and only if the barycenter of P is the origin. Throughout, dP_k denotes the del Pezzo surface obtained by blowing up $\mathbb{P}^2$ in k points, torus-fixed for $k \leq 3$; thus dP_3 is the smooth toric surface whose ray generators are the six roots of A_2 .

2.2. Klyachko data of the tangent bundle. An equivariant reflexive sheaf on X_Σ is equivalent to a finite collection of decreasing filtrations $\{E^\rho(j)\}_{j \in \mathbb{Z}}$ of a fixed vector space E , one for each ray [Kly90, Per04, Koo11]. For the tangent bundle, $E = N_\mathbb{C}$ and the filtrations

are two-step:

$$E^\rho(j) = \begin{cases} N_{\mathbb{C}}, & j \leq 0, \\ \mathbb{C} u_\rho, & j = 1, \\ 0, & j \geq 2. \end{cases}$$

For a subspace $V \subseteq N_{\mathbb{C}}$, the associated saturated equivariant subsheaf $\mathcal{F}_V \subseteq T_X$ has filtrations $V \cap E^\rho(j)$, whence by (1)

$$(2) \quad \deg_{-K} \mathcal{F}_V = \sum_{\rho: u_\rho \in V} \text{Vol}(F_\rho), \quad \mu(\mathcal{F}_V) = \frac{\deg_{-K} \mathcal{F}_V}{\dim V}.$$

Proposition 2.1 (Klyachko; Kool; [HNS22, Prop. 1.2]). *Let X be a smooth toric Fano n -fold. Then T_X is stable (resp. semistable) with respect to $-K_X$ if and only if for every subspace $0 \neq V \subsetneq N_{\mathbb{C}}$ spanned by ray generators,*

$$\frac{1}{\dim V} \sum_{\rho: u_\rho \in V} \text{Vol}(F_\rho) < \frac{(-K_X)^n}{n} \quad (\text{resp. } \leq).$$

Two reductions underlie the statement, and we record them because the polystability refinement reuses both. By the equivariant stability criterion for toric sheaves [Koo11, HNS22], both stability and semistability may be tested on saturated equivariant subsheaves. For instability this follows immediately from uniqueness of the maximal Harder–Narasimhan subsheaf; the equal-slope boundary needed for stability is the corresponding equivariant socle statement. Thus it is enough to test subspaces $V \subseteq N_{\mathbb{C}}$. Second:

Lemma 2.2. *For any subspace $V \subseteq N_{\mathbb{C}}$ containing at least one ray generator, let $V' = \text{span}\{u_\rho : u_\rho \in V\}$. Then $\mu(\mathcal{F}_{V'}) \geq \mu(\mathcal{F}_V)$.*

Proof. $V' \subseteq V$ contains exactly the same ray generators, so the numerator of (2) is unchanged while $\dim V' \leq \dim V$. $\square$

Consequently it suffices to range over spans of linearly independent subsets of ray generators, of which there are finitely many; and every quantity in Proposition 2.1 is an integer or a rational number computed exactly from the polytope. Stability of T_X on a smooth toric Fano is therefore decidable by exact arithmetic, and the decision comes with a proof: either the full list of subspace slopes (for stability), or an explicit destabilizing subspace (for instability).

3. THE COMPUTATION AND ITS VALIDATION

3.1. Algorithm. For each variety, presented by its ray generators: (i) enumerate the vertices of the dual polytope P exactly, verifying integrality (reflexivity) along the way; (ii) compute each facet volume $\text{Vol}(F_\rho)$ by an exact recursive triangulation of the facet in a lattice basis of its supporting hyperplane; (iii) enumerate the ray-spanned subspaces V (deduplicated by reduced row echelon form over $\mathbb{Q}$) with $1 \leq \dim V \leq n-1$, and compare each slope (2) with $(-K)^n/n$; (iv) record the maximal subsheaf slope, all equality witnesses, and the exact barycenter of P . In an unstable case the stored witness is the sum of all subspaces attaining the maximal slope, hence the first Harder–Narasimhan subspace rather than an enumeration-dependent maximizer. All arithmetic is over $\mathbb{Z}$ and $\mathbb{Q}$ (fraction-free Bareiss elimination for determinants); no floating point enters any verdict. Every variety is required to pass the internal identity $\sum_\rho \text{Vol}(F_\rho) = n! \text{vol}(P)$ of (1), with the right-hand side obtained by coning the computed facet triangulations to the origin. This is a consistency identity,

not an independent certification of the triangulations. External checks are supplied by the published-result and database comparisons below.

For the dimension-7 and 8 members of the family of Section 7, which lie beyond the complete census above, the reported degrees, slopes and barycenters are also evaluated directly from the exact hexagon formulas of Lemma 7.4 and Proposition 7.6: the relevant polynomials are integrated over the fixed rational hexagon and its edges in rational arithmetic. An accelerated full-dimensional implementation supplies an auxiliary cross-check. It uses floating point only to propose vertex bases and Delaunay simplices, then evaluates retained vertices and simplex determinants exactly. For the reported inputs, an exhaustive exact basis audit confirmed that the proposal stage omitted no true vertex. The accelerated and reference implementations agree on every degree and verdict wherever both run (dimensions ≤ 6), and both agree with the exact hexagon computation.

3.2. Data. The classification data (vertex sets of all smooth Fano polytopes in dimensions 3–6) were obtained from the `polyDB` database (collection `Polytopes.Lattice.SmoothReflexive`, data of Paffenholz computed by Øbro’s algorithm [Øbr07, Paf17]), stored there in the dual convention and converted exactly (each facet normal $[1, a]$ of the stored polytope contributes the vertex a of the fan polytope; no floating-point dualization). During ingestion every polytope was re-verified as smooth and reflexive by exact arithmetic: all vertices are primitive and every facet is a unimodular simplex at lattice distance one. A validation script distributed with the code reconstructs every facet from the ray data and checks these conditions, including that every listed ray is a genuine vertex; the imported files also record the conversion protocol in their metadata. Counts match the classification: 18, 124, 866, 7622. For dimensions 3 and 5 the vertex data were additionally compared file-by-file against Paffenholz’s original distribution and found identical. In dimensions 3 and 4, where the imported `polyDB` metadata include an independently stored lattice volume, all 142 values agree with the computed anticanonical degrees.

3.3. Validation against published results. The implementation reproduces every published result that overlaps its range:

- (1) *Surfaces* (Fahlaoui [Fah89]; [HNS22, Thm. 1.3, Cor. 3.1]): $\mathbb{P}^2$, dP_2 , dP_3 stable; $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{F}_1$ strictly semistable, including both equality cases, which test the semistable boundary of the criterion.
- (2) *Fano three-folds* (Steffens [Ste96]): of the 18 toric cases our computation finds 7 unstable and 7 strictly semistable; these agree exactly with the toric members of Steffens’s lists, including $\text{Bl}_\ell \mathbb{P}^3$ and the del Pezzo products among the strictly semistable cases.
- (3) *Four-folds of Picard rank ≤ 3* ([BDGP21]; [DDK20, Table 1]): all 38 varieties were matched to our database entries by exact $GL(4, \mathbb{Z})$ -equivalence of fans (necessary, since seven values of $(-K)^4$ are each shared by two to four varieties), and all 38 verdicts agree, as do the slopes $\mu(T_X)$ reported in the proofs of [DDK20].
- (4) *Kähler–Einstein tables* ([Mab87, Nak93, Nak94, NSY23]): our barycenter column reproduces the known lists of Kähler–Einstein toric Fanos in dimensions ≤ 4 ; in particular the rank- ≤ 3 four-fold list $\{\mathbb{P}^4, B_4, C_4, D_{13}\}$.
- (5) *Hand-checked cases*: $\mathbb{P}^n$, products, and projective bundles in dimensions 2–4, where all degrees and slopes are classical.

One near-miss shows why the matching in item (3) is necessary: verdicts cannot be transported by numerical invariants alone, and [NSY23] tabulate *Ding* stability, a different notion;

three of the rank-3 four-folds (G_4 – G_6) are Ding-unstable with slope-stable tangent bundle. No contradiction is involved.

4. THE CLASSIFICATION

Theorem 4.1. *The $(-K)$ -slope stability of the tangent bundle of every smooth toric Fano variety of dimension $3 \leq n \leq 6$ is as follows.*

n	total	stable	strictly semistable	unstable
3	18	4	7	7
4	124	26	24	74
5	866	109	82	675
6	7622	636	366	6620

Proof. By Proposition 2.1 and Lemma 2.2, stability is decided by the finitely many exact slope comparisons of Section 3, executed in integer and rational arithmetic on the classified polytopes [Øbr07, Paf17]; Section 3.3 records the validation. $\square$

Complete tables (for each variety the database identifier, ray data, facet degrees, the exact extremal slopes, a subspace attaining the maximal slope, and the stability type) accompany the paper as ancillary files and are also available, together with all code and input data, in the public repository cited in the introduction.

The distribution by Picard rank exhibits the two phenomena announced in the introduction: the dominance of instability, and the collapse of stability at high rank. (In the tables, st, sss and un abbreviate stable, strictly semistable and unstable.)

$n = 4$					$n = 5$				
ρ	#	st	sss	un	#	st	sss	un	
1	1	1	0	0	1	1	0	0	
2	9	0	3	6	15	0	3	12	
3	28	6	3	19	91	9	5	77	
4	47	9	8	30	268	34	13	221	
5	27	7	4	16	312	46	23	243	
6	10	3	4	3	137	18	18	101	
7	1	0	1	0	35	1	15	19	
8	1	0	1	0	5	0	3	2	
9	–	–	–	–	2	0	2	0	

and for $n = 6$ (all 7622 six-folds, with ρ up to 12) the distribution is:

ρ	#	st	sss	un	ρ	#	st	sss	un
1	1	1	0	0	7	771	66	69	636
2	26	0	4	22	8	186	10	46	130
3	257	13	7	237	9	39	0	17	22
4	1318	85	30	1203	10	11	0	8	3
5	2807	246	75	2486	11	1	0	1	0
6	2204	215	108	1881	12	1	0	1	0

The rows $\rho \leq 3$ for $n = 4$ coincide with [BDGP21, DDK20]; the rows $\rho \geq 4$ (86 varieties) and the whole of $n = 5, 6$ are new. Stability collapses at high rank in every dimension: the last stable examples occur at $\rho = 6$ ($n = 4$), $\rho = 7$ ($n = 5$) and $\rho = 8$ ($n = 6$), and above

these ranks the varieties (pseudo-symmetric ones and products) are never stable (a product tangent bundle splits). The unique stable five-fold with $\rho = 7$ is the subject of Section 7; its higher-dimensional analogues populate the last stable rank in each dimension.

5. POLYSTABILITY AND HERMITIAN–EINSTEIN METRICS

Semistability does not decide the existence of a Hermitian–Einstein metric; polystability does [Don85, UY86]. Polystability of an equivariant sheaf can, a priori, be exhibited by a non-equivariant splitting; the first point is that on a toric variety it never has to be (compare [Kly90, Koo11]).

Lemma 5.1. *Let X be a smooth projective toric variety with torus T and let E be a T -equivariant reflexive sheaf that is polystable with respect to a fixed polarization. Then E admits a T -equivariant decomposition into T -equivariant stable subsheaves of equal slope.*

Proof. Write $E \cong \bigoplus_j S_j^{\oplus a_j}$ with the S_j stable of equal slope and pairwise non-isomorphic. For $t \in T$ we have $t^*E \cong E$; torus translations preserve the polarization, so each t^*S_j is again a stable summand of E of the same slope and its isomorphism class lies in the finite set $\{[S_1], \dots, [S_r]\}$. Each locus $\{t \in T : t^*S_j \cong S_{j'}\}$ is closed (it is where $\dim \operatorname{Hom}(t^*S_j, S_{j'}) \geq 1$, and a nonzero map between stable sheaves of equal slope is an isomorphism, so these loci are disjoint); finitely many disjoint closed sets covering the irreducible variety T force one of them to be everything, and it contains $t = 1$ for $j' = j$. Hence $t^*S_j \cong S_j$ for all t . The isotypic summand is canonically

$$B_j = \operatorname{im}(S_j \otimes \operatorname{Hom}(S_j, E) \longrightarrow E).$$

It therefore satisfies $t^*B_j = B_j$, is an equivariant subsheaf, and $E = \bigoplus_j B_j$ equivariantly.

Next, each $S = S_j$ itself carries an equivariant structure. The pairs (t, ψ) with $\psi: t^*S \xrightarrow{\sim} S$ form an affine algebraic group $\tilde{T}$, a central extension $1 \rightarrow \mathbb{G}_m \rightarrow \tilde{T} \rightarrow T \rightarrow 1$ because S is simple; S is tautologically $\tilde{T}$ -equivariant. Every such extension of a torus splits: the commutator pairing $T \times T \rightarrow \mathbb{G}_m$ is bimultiplicative, so $t \mapsto e(t, \cdot)$ is a morphism from the connected group T to the discrete character lattice, hence trivial; $\tilde{T}$ is therefore a commutative extension of tori, hence a torus, and the extension splits because its character sequence $0 \rightarrow X^*(T) \rightarrow X^*(\tilde{T}) \rightarrow X^*(\mathbb{G}_m) \rightarrow 0$ has free quotient $X^*(\mathbb{G}_m) \cong \mathbb{Z}$. A splitting $T \rightarrow \tilde{T}$ makes S itself T -equivariant.

Finally $W_j = \operatorname{Hom}(S_j, B_j)$ is a finite-dimensional T -representation, and evaluation $S_j \otimes W_j \rightarrow B_j$ is an equivariant isomorphism; decomposing W_j into characters χ decomposes B_j equivariantly into the sheaves $S_j \otimes \chi$, each equivariant with underlying sheaf S_j , hence stable of the same slope. $\square$

For toric tangent bundles the equivariant question is then purely combinatorial, because of the following elementary observation.

Proposition 5.2. *Let X be a smooth toric variety. Equivariant direct-sum decompositions $T_X \cong \bigoplus_{i=1}^m \mathcal{F}_{V_i}$ correspond exactly to decompositions $N_{\mathbb{C}} = \bigoplus_{i=1}^m V_i$ such that every ray generator u_ρ lies in some V_i . If X is in addition projective and Fano, then T_X is polystable with respect to $-K_X$ if and only if the ray generators admit a partition into groups $\Gamma_1, \dots, \Gamma_m$ whose spans $V_i = \operatorname{span} \Gamma_i$ are linearly independent with $\sum_i \dim V_i = n$, such that each $\mathcal{F}_{V_i}$ has slope $(-K)^n/n$ and each is stable, i.e. every ray-spanned proper subspace $0 \neq U \subsetneq V_i$ satisfies $\mu(\mathcal{F}_U) < (-K)^n/n$.*

Proof. A direct sum of equivariant reflexive sheaves has, on each ray, the direct sum of the filtrations. Since the filtrations of T_X are two-step, a decomposition $N_{\mathbb{C}} = \bigoplus V_i$ is compatible with them if and only if for every ρ the line $\mathbb{C}u_\rho$ equals $\bigoplus_i (V_i \cap \mathbb{C}u_\rho)$, i.e. $u_\rho \in V_i$ for some (necessarily unique) i ; in that case the summand with fiber V_i has exactly the filtrations of $\mathcal{F}_{V_i}$, and Klyachko data determine the sheaf. The polystability statement follows since a polystable bundle is a direct sum of stable sheaves of equal slope, the decomposition may be taken equivariant by Lemma 5.1, and stability of each summand is tested by the same equivariant stability criterion as above and then, by Lemma 2.2, on ray-spanned subspaces of V_i . $\square$

The splitting has a canonical matroid-theoretic and geometric form.

Corollary 5.3 (Ray-matroid and product criterion). *Let $X = X_\Sigma$ be a smooth toric Fano variety, and let M_Σ be the linear matroid represented by the ray generators. Write $C_1, \dots, C_s$ for its connected components, where two rays are equivalent if they can be joined by a chain of minimal linear dependences, and put*

$$V_a = \text{span}_{\mathbb{C}}\{u_\rho : \rho \in C_a\}.$$

Then $N_{\mathbb{C}} = \bigoplus_a V_a$, and this is the finest decomposition compatible with all ray lines. It is induced by a product decomposition

$$X \cong \prod_{a=1}^s X_a, \quad T_X \cong \bigoplus_{a=1}^s p_a^* T_{X_a},$$

where each X_a is a smooth toric Fano variety with fan in $N \cap V_a$. Every other equivariant direct-sum decomposition of T_X is obtained by grouping some of these factors.

If T_X is semistable, every component summand has slope $\mu(T_X)$ and is semistable. Moreover, T_X is polystable if and only if every T_{X_a} is $(-K_{X_a})$ -stable. Consequently, a strictly semistable T_X is polystable if and only if X is a nontrivial product of smooth toric Fano varieties with stable tangent bundles.

Proof. A circuit cannot meet two summands of a direct-sum decomposition containing every ray in one summand: projecting its dependence to either summand would give a dependence on a proper subset of the circuit. Thus every C_a lies in one summand of every decomposition in Proposition 5.2. Conversely, a linear dependence meeting two components, chosen with minimal support, would be such a circuit. Hence the V_a are linearly independent; since the rays span $N_{\mathbb{C}}$, their direct sum is all of $N_{\mathbb{C}}$. This proves the canonical splitting and its fineness.

Choose a maximal cone of Σ . Its primitive generators form a lattice basis, and their subsets in the spaces V_a show that $N = \bigoplus_a (N \cap V_a)$. Every cone of Σ is the direct sum of its intersections with the V_a . Because intersections and faces of cones are componentwise under this direct-sum decomposition, the collections $\Sigma_a = \{\sigma \cap V_a : \sigma \in \Sigma\}$ are fans. They are complete: apply completeness of Σ to a point of V_a . Moreover every combination of maximal cones of the Σ_a occurs in Σ . Indeed, choose an interior point in each and apply completeness to their sum; uniqueness of the component decomposition identifies the resulting intersections. Thus $\Sigma = \prod_a \Sigma_a$ and $X = \prod_a X_a$. Each factor is smooth and Fano, and its tangent pullback is the summand $\mathcal{F}_{V_a}$.

Set $n_a = \dim X_a$ and $d_a = (-K_{X_a})^{n_a}$. For an equivariant subsheaf $\mathcal{G} \subseteq T_{X_a}$, the product intersection formula gives

$$\deg_{-K_X}(p_a^*\mathcal{G}) = \frac{(n-1)!}{(n_a-1)! \prod_{b \neq a} n_b!} \left(\prod_{b \neq a} d_b \right) \deg_{-K_{X_a}}(\mathcal{G}).$$

By the Klyachko filtrations, every saturated equivariant subsheaf of $p_a^*T_{X_a}$ used in the stability test is the pullback of the corresponding subsheaf of T_{X_a} . Thus the strict slope comparisons inside the a -th summand are exactly those for T_{X_a} , while

$$\mu_{-K_X}(p_a^*T_{X_a}) = \frac{(n-1)!}{\prod_b n_b!} \prod_b d_b$$

is independent of a . Consequently, stable factor tangents give a polystable T_X .

Suppose now that T_X is semistable. Each component summand has slope at most $\mu(T_X)$; since their ranks and degrees add to those of T_X , every component slope equals $\mu(T_X)$. Any subsheaf destabilizing a component would then destabilize T_X , so every component is semistable.

Conversely, suppose T_X is polystable. Lemma 5.1 supplies an equivariant decomposition into stable equal-slope summands. By fineness, each is a union of matroid components, but it cannot contain two: either component would be a proper subsheaf of the same slope. Hence the stable summands are precisely the component summands, and the factor tangent bundles are stable by the scaling formula above. $\square$

The census was computed by the finite search over ray partitions of Proposition 5.2; Corollary 5.3 gives an equivalent canonical test using the connected components of the ray matroid. The two tests agree on every strictly semistable case. Applying them to Theorem 4.1 gives:

Theorem 5.4. *Among the strictly semistable varieties, exactly 4 of 7 ($n = 3$), 14 of 24 ($n = 4$), 52 of 82 ($n = 5$), and 259 of 366 ($n = 6$) are polystable. Combined with the stable count, the tangent bundle admits a Hermitian–Einstein metric with respect to a Kähler form representing $c_1(X)$ for exactly*

$$8 \text{ of } 18 \text{ } (n = 3), \quad 40 \text{ of } 124 \text{ } (n = 4), \quad 161 \text{ of } 866 \text{ } (n = 5), \quad 895 \text{ of } 7622 \text{ } (n = 6)$$

smooth toric Fano varieties.

Proof. By Lemma 5.1, Proposition 5.2, and Corollary 5.3, polystability of each strictly semistable T_X is decided by the equivalent splitting tests above, executed in exact arithmetic; combining the resulting counts with the stable counts of Theorem 4.1 and the Kobayashi–Hitchin correspondence [Don85, UY86] gives the totals. $\square$

Example 5.5 (Semistable but not polystable). The semistable-but-not-polystable class is the exact locus where semistability fails to deliver a metric, and its members are structured:

- (1) The blow-up $\text{Bl}_\ell \mathbb{P}^3$ of $\mathbb{P}^3$ along a line is strictly semistable but not polystable: the three-dimensional analogue of the Hirzebruch surface $\mathbb{F}_1$, whose equality subsheaf is the relative tangent sheaf of the $\mathbb{P}^1$ -fibration structure.
- (2) $\mathbb{F}_1 \times \mathbb{P}^1$ is a *product* that is not polystable: its equality decomposition requires the factor bundle $T_{\mathbb{F}_1}$, which is itself only strictly semistable. Products of semistables are semistable, but polystability demands stable summands.
- (3) By contrast $d\mathbb{P}_2 \times \mathbb{P}^1$ is polystable ($T_{d\mathbb{P}_2}$ is stable by [Fah89], and the slopes balance), so it carries a Hermitian–Einstein tangent bundle although it has no Kähler–Einstein metric.

6. COMPARISON WITH THE KÄHLER–EINSTEIN CONDITION

For toric Fanos the Kähler–Einstein condition is the vanishing of the barycenter of P [Mab87, WZ04], which we compute exactly for every variety. The comparison runs in two directions.

Kähler–Einstein $\Rightarrow$ *polystable*. This is the Kobayashi–Lübke direction [Kob87, Lüb83], and the data confirm it without exception: no Kähler–Einstein variety in the census is unstable, and every strictly semistable Kähler–Einstein variety is polystable (3/3 in dimension 3, 8/8 in dimension 4, 16/16 in dimension 5, 38/38 in dimension 6). We emphasize that this is a useful end-to-end comparison with the independently published Kähler–Einstein classifications in low dimensions. Internally, the barycenter and slope tests are different functions of the same exact polytope data. It is also the effective form of the subspace concentration conditions of [HNS22, HL14]; for their affine refinements see [Wu23, FHK23].

Polystable $\nRightarrow$ *Kähler–Einstein*, *overwhelmingly in the census*. Fahlaoui’s example dP_2 is the smallest instance of what our census shows to be the dominant behaviour:

n	stable	of which KE	stable non-KE
2	3	2	1
3	4	2	2
4	26	4	22
5	109	7	102
6	636	13	623

In dimension five, 94% of the varieties with stable tangent bundle admit no Kähler–Einstein metric, and in dimension six the fraction rises to 98% (623 of 636): the Kähler–Einstein proportion of the stable locus continues to decay with dimension. A Hermitian–Einstein metric on T_X with respect to an anticanonical Kähler form is thus a far weaker, and far more common, Einstein condition than a Kähler–Einstein metric on X , already within the toric world. Counting stable and polystable tangent bundles together, the number of varieties with this Hermitian–Einstein metric but without a Kähler–Einstein metric is 3, 28, 138, and 844 in dimensions 3, 4, 5, and 6, respectively.

7. THE ROOT-TWISTED dP_3 -FIBRATIONS

Among the 42 smooth toric Fano five-folds of Picard rank ≥ 7 , exactly one, polyDB entry F.5D.0611, has stable tangent bundle. It has 12 rays, 48 maximal cones, $(-K)^5 = 2580$, facet degrees $(278 \times 6, 156 \times 3, 148 \times 3)$, and $\mu(T_X) = 516$ against a maximal subsheaf slope of 506; its barycenter is zero, so it is Kähler–Einstein. Its structure is revealed by its primitive collections, all of cardinality two: six of its rays lie in a rank-two sublattice, forming the A_2 root hexagon $\{\pm\alpha, \pm\beta, \pm(\alpha+\beta)\}$ (the fan of dP_3), and the projection along that plane maps the remaining six rays onto the fan of $(\mathbb{P}^1)^3$. The fibration is not a product: the three base directions lift with twists $\alpha, \beta, -(\alpha+\beta)$ in the fiber plane.

Definition 7.1. Write $R = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta)\}$ for the six roots, where α, β is a basis of a rank-two lattice N_f . Fix $k \geq 0$ and twists $t_1, \dots, t_k \in \{0\} \cup R$, and let f_i be the i -th standard basis vector of $\mathbb{Z}^k$. Put $u_i^+ = f_i + t_i$ and $u_i^- = -f_i$. In $N_f \oplus \mathbb{Z}^k$, let $\Sigma(t_\bullet)$ be the fan whose maximal cones are

$$\mathrm{cone}(\rho, \rho', u_1^{\varepsilon_1}, \dots, u_k^{\varepsilon_k}), \quad \varepsilon_i \in \{+, -\},$$

where ρ, ρ' run over adjacent rays of the hexagonal fan of dP_3 . Define $X(t_1, \dots, t_k) = X_{\Sigma(t_\bullet)}$, of dimension $n = 2 + k$. Lemma 7.4 below verifies directly that these cones form the smooth complete normal fan of a reflexive polytope.

Each $X(t_1, \dots, t_k)$ is a smooth toric variety fibered over $(\mathbb{P}^1)^k$ with fiber dP_3 ; the untwisted case is $X(0, \dots, 0) = \mathrm{dP}_3 \times (\mathbb{P}^1)^k$. Every $X(t_\bullet)$ is in fact a smooth toric *Fano* variety of Picard rank $n + 2$ (Lemma 7.4 below). The variety above is $X(\alpha, \beta, -(\alpha + \beta))$: its twist multiset is a full orbit of the order-three rotation $\tau: \alpha \mapsto \beta \mapsto -(\alpha + \beta)$ of the root hexagon, a symmetry that extends to an automorphism of the variety and forces the barycenter of the moment polytope to the origin (see the proof of Theorem 7.11), so X is Kähler–Einstein.

The computations to date are all exact: dimensions ≤ 6 use the reference implementation of Section 3, and every row is independently reproduced by the rational hexagon formulas below.

twists	n	$(-K)^n$	stability	KE	remark
$(\alpha, -\alpha)$	4	268	stable	yes	the smallest zero-sum pair
(α, β)	4	278	stable	no	the smallest unbalanced stable example
$(\alpha, \beta, -(\alpha + \beta))$	5	2580	stable	yes	= F.5D.0611, the $\rho \geq 7$ example
$(0, 0, 0)$	5	2880	strictly ss	yes	the product
$(\alpha, -\alpha, \beta, -\beta)$	6	29928	stable	yes	zero-sum pairs
$(\alpha, \beta, -(\alpha + \beta), 0)$	6	30960	strictly ss	yes	one untwisted factor
$(\alpha, \beta, -(\alpha + \beta), \alpha)$	6	30960	strictly ss	no	twist sum $\alpha \neq 0$
$(\alpha, \beta, -(\alpha + \beta), \alpha, -\alpha)$	7	404544	stable	no	zero-sum, no fiber symmetry
$(\alpha, \beta, -(\alpha + \beta))^{\times 2}$	8	6242544	stable	yes	two τ -orbits

Theorem 7.2 (zero-sum stability). *Let $t_1, \dots, t_k \in R$ be nonzero roots with $\sum_i t_i = 0$. Then $T_{X(t_\bullet)}$ is $(-K)$ -stable. In particular, smooth toric Fano varieties of Picard rank $n + 2$ with stable tangent bundle exist in every dimension $n \geq 4$.*

Theorem 7.3 (Kähler–Einstein classification). *Among the zero-sum cases of Theorem 7.2, $X(t_\bullet)$ is Kähler–Einstein if and only if the twist multiset $\{t_1, \dots, t_k\}$ is invariant under negation $t \mapsto -t$ or under the order-three rotation $\tau: \alpha \mapsto \beta \mapsto -(\alpha + \beta)$ of the root hexagon.*

Two mechanisms underlie the statements. *Stability*: an untwisted factor leaves an equal-slope summand, while for a balanced collection of nonzero root twists every proper ray-spanned subspace generates a subsheaf of slope strictly below $\mu(T_X)$ (Theorem 7.2). The twist is thus a geometric counterpart of the stable perturbations of semistable tangent sheaves that Napame and Tipler obtain by varying the polarization or blowing up [NT24]: here the polarization remains anticanonical, and it is the fibration itself that is deformed. *The Kähler–Einstein criterion*: whenever the twist multiset is preserved by negation or by the rotation τ , that symmetry extends to an automorphism of $X(t_\bullet)$, permuting the base factors. The barycenter of the moment polytope is fixed by the induced map, whose action on the fibre component is negation resp. the rotation (neither fixes a nonzero vector), so the fibre component of the barycenter vanishes; and the base components vanish with it, being proportional to it (Lemma 7.4(3) below). This proves sufficiency; necessity follows from the extreme-exponent ordering in Lemma 7.10 below. The statement also corrects the naive expectation that a vanishing twist *sum* alone forces the barycenter to zero. The minimal counterexample to that expectation is the dimension-seven variety $X(\alpha, \beta, -(\alpha + \beta), \alpha, -\alpha)$: its twists are nonzero roots summing to zero, but the multiset is invariant under neither symmetry, and

the variety is $(-K)$ -stable with a *nonzero* barycenter: a stable, non-Kähler–Einstein tangent bundle beyond the complete census carried out here.

For comparison, exact computation through dimension eight gives nine zero-sum classes of nonzero roots up to the hexagon’s dihedral symmetry and reproduces Theorems 7.2 and 7.3 in every case (τ -invariance and negation-invariance are marked $\checkmark$):

n	twist multiset	neg-inv.	τ -inv.	Kähler–Einstein
4	$\alpha, -\alpha$	$\checkmark$		yes
5	$\alpha, \beta, -(\alpha+\beta)$		$\checkmark$	yes
6	$\alpha, -\alpha, \alpha, -\alpha$	$\checkmark$		yes
6	$\alpha, \beta, -\alpha, -\beta$	$\checkmark$		yes
7	$\alpha, \alpha, \beta, -\alpha, -(\alpha+\beta)$			no
8	$\alpha, \alpha, \alpha, -\alpha, -\alpha, -\alpha$	$\checkmark$		yes
8	$\alpha, \alpha, \beta, \beta, -(\alpha+\beta), -(\alpha+\beta)$		$\checkmark$	yes
8	$\alpha, \alpha, \beta, -\alpha, -\alpha, -\beta$	$\checkmark$		yes
8	$\alpha, \beta, \alpha+\beta, -\alpha, -\beta, -(\alpha+\beta)$	$\checkmark$	$\checkmark$	yes

Kähler–Einstein holds in exactly the eight classes with a symmetry and fails in the single class (dimension seven) with neither.

The prism structure and the stability criterion. The moment polytopes of the family are explicit enough to begin the proof of Theorem 7.2. Write $Q \subset M_f \otimes \mathbb{Q}$, with M_f the lattice dual to N_f , for the hexagon $\{x : \langle x, \rho \rangle \geq -1 \text{ for all six roots } \rho\}$ (the anticanonical polytope of dP_3 ; lattice area 3, six edges $E_\rho \subset \{\langle x, \rho \rangle = -1\}$ of lattice length one), set $\ell_t(x) = \langle x, t \rangle$, and

$$F(x) = \prod_{i=1}^k (2 + \ell_{t_i}(x)).$$

Lemma 7.4 (prism structure). *Let $t_1, \dots, t_k \in \{0\} \cup R$ and $n = 2 + k$.*

- (1) *$X(t_1, \dots, t_k)$ is a smooth toric Fano variety of Picard rank $n + 2$, with anticanonical polytope the generalized prism*

$$P = \{(x, y) \in (M_f \oplus \mathbb{Z}^k) \otimes \mathbb{Q} : x \in Q, -1 - \ell_{t_i}(x) \leq y_i \leq 1\},$$

whose fibre over $x \in Q$ is a box with side lengths $2 + \ell_{t_i}(x) \in [1, 3]$.

- (2) *The anticanonical degrees are hexagon integrals:*

$$(-K)^n = n! \int_Q F, \quad \deg D_{f_i+t_i} = \deg D_{-f_i} = (n-1)! \int_Q \frac{F}{2 + \ell_{t_i}}, \quad \deg D_\rho = (n-1)! \int_{E_\rho} F,$$

with the lattice measure on E_ρ ; in particular $\mu(T_X) = (n-1)! \int_Q F$.

- (3) *The barycenter $b = (b_x, b_y)$ of P has fibre component $b_x = \int_Q xF / \int_Q F$ and base components $b_{y_i} = -\frac{1}{2} \langle b_x, t_i \rangle$. In particular $X(t_\bullet)$ is Kähler–Einstein if and only if the hexagon moment $\int_Q x F(x) dx$ vanishes.*

Proof. (1) The inequalities $\langle (x, y), u \rangle \geq -1$, over the rays of Definition 7.1, read $x \in Q$ for the six roots and $-1 - \ell_{t_i}(x) \leq y_i \leq 1$ for the pair $f_i + t_i, -f_i$; this is P . Its vertices are integral: over a vertex v of Q the box endpoints are 1 and $-1 - \ell_{t_i}(v)$ with $\ell_{t_i}(v) \in \{0, \pm 1\}$, and the two never coincide since the widths lie in $[1, 3]$. Every defining inequality supports a facet (a prism over an edge E_ρ for the root inequalities; the graph of an affine function over the prism with the i -th factor omitted, for the base inequalities), so P is reflexive with

the rays of Definition 7.1 as its facet normals. The normal cone at each of the $6 \cdot 2^k$ vertices is spanned by two adjacent hexagon roots together with one of $f_i + t_i$, $-f_i$ per factor, a unimodular basis; hence the normal fan of P is the smooth complete fan of Definition 7.1 and $X(t_\bullet)$ is Fano, of Picard rank $(6 + 2k) - n = n + 2$. (2) Integrating the box volume gives $\text{vol}(P)$, hence $(-K)^n$ by (1). The facet $\{y_i = 1\}$ projects isomorphically (dropping y_i) to the prism with the i -th factor omitted, and the facet $\{y_i = -1 - \ell_{t_i}\}$ is carried to it by the unimodular shear $y_i \mapsto -y_i - \ell_{t_i}(x)$, giving the base degrees; the facet over E_ρ is the same prism construction over the edge. All identifications are lattice isomorphisms, so normalized volumes agree. (3) The y_i -fibre over x is the interval $[-1 - \ell_{t_i}(x), 1]$, with midpoint $-\ell_{t_i}(x)/2$; hence $b_{y_i} = -\frac{1}{2} \int_Q \ell_{t_i} F / \int_Q F = -\frac{1}{2} \langle b_x, t_i \rangle$ by linearity of ℓ_{t_i} . So $b = 0$ if and only if $b_x = 0$, and the Kähler–Einstein claim is Wang–Zhu [Mab87, WZ04]. $\square$

The comparisons of Proposition 2.1 now become inequalities between hexagon integrals. The bridge is the divergence theorem applied to the field xF on the reflexive polygon Q : since every facet of Q lies on $\{\langle x, \rho \rangle = -1\}$ with ρ primitive, the Euclidean boundary factors cancel and

$$(3) \quad \int_{\partial Q} F = 2 \int_Q F + \int_Q \langle x, \nabla F \rangle, \quad \langle x, \nabla F \rangle = F \sum_{i=1}^k \frac{\ell_{t_i}}{2 + \ell_{t_i}},$$

with the boundary integral taken in the lattice measure. Writing $\Phi(t_\bullet) = \int_Q F \sum_i \ell_{t_i} / (2 + \ell_{t_i})$, the fibre subsheaf $\mathcal{F}_{N_f}$ (the relative tangent sheaf of $\pi: X \rightarrow (\mathbb{P}^1)^k$, and the subsheaf whose slope equality makes the untwisted product merely polystable) satisfies, by Lemma 7.4 and (3),

$$(4) \quad \mu(\mathcal{F}_{N_f}) - \mu(T_X) = \frac{(n-1)!}{2} \Phi(t_\bullet).$$

Call a proper ray-spanned subspace V , or the subsheaf $\mathcal{F}_V$, *critical* if $\mu(\mathcal{F}_V) = \mu(T_X)$ and *subcritical* if $\mu(\mathcal{F}_V) < \mu(T_X)$; by Proposition 2.1, stability of T_X is the statement that every proper ray-spanned subspace is subcritical.

Proposition 7.5. *Assume $k \geq 1$ and let every t_i lie in $\{0\} \cup R$.*

- (1) *If some $t_i = 0$, then $\mu(\mathcal{F}_{\mathbb{C}f_i}) = \mu(T_X)$ exactly; hence T_X is not stable.*
- (2) *If the t_i are all equal to a single nonzero root, then $\Phi \geq 0$, strictly for $k \geq 2$; hence T_X is not stable, and is unstable for $k \geq 2$.*
- (3) *If the t_i are nonzero roots with $\sum_i t_i = 0$, then $\Phi < 0$: the fibre subsheaf is subcritical.*

Proof. (1) With $t_i = 0$ the line $V = \mathbb{C}f_i$ contains exactly the two rays $\pm f_i$, each of degree $(n-1)! \int_Q F/2$ by Lemma 7.4, so $\mu(\mathcal{F}_V) = (n-1)! \int_Q F = \mu(T_X)$. (2) With all $t_i = t$, $\Phi = k \int_Q (2 + \ell_t)^{k-1} \ell_t$; folding the centrally symmetric Q along $x \mapsto -x$ gives $\Phi = k \int_{Q \cap \{\ell_t > 0\}} \ell_t [(2 + \ell_t)^{k-1} - (2 - \ell_t)^{k-1}] \geq 0$, strict for $k \geq 2$; conclude by (4). (3) The function $u \mapsto u/(2 + u)$ is strictly concave for $u > -2$, so pointwise on Q , by Jensen,

$$\sum_i \frac{\ell_{t_i}(x)}{2 + \ell_{t_i}(x)} \leq k \frac{\bar{\ell}(x)}{2 + \bar{\ell}(x)}, \quad \bar{\ell} = \frac{1}{k} \ell_{\sum_i t_i} = 0,$$

with equality only where all $\ell_{t_i}(x)$ coincide. A zero-sum multiset of nonzero roots contains two distinct roots, so the equality locus is contained in a proper linear subspace and has measure zero. Hence $\Phi < 0$ strictly since $F \geq 1$ on Q . $\square$

Proposition 7.5(1) proves that an untwisted factor always obstructs stability, while (2) gives a simple class of unbalanced unstable examples. Part (3) settles, in the zero-sum case, the slope inequality of the subsheaf that dominates the statistics of Section 8. In fact the entire stability question reduces to finitely many hexagon integrals, in a form that proves Theorem 7.2 for every zero-sum multiset. For the i -th twist define its *singleton integral* $\varphi_i = \int_Q F \ell_{t_i} / (2 + \ell_{t_i})$ (so $\Phi = \sum_i \varphi_i$), and for each of the three root lines $\{\pm\rho\}$ define the *line share*

$$S_\rho = \int_{E_\rho} F + \int_{E_{-\rho}} F + \sum_{i: t_i = \pm\rho} \max(0, -\varphi_i).$$

Proposition 7.6 (reduction). *Let all t_i be nonzero roots. Then T_X is $(-K)$ -stable if and only if*

- (a) $\varphi_i < 0$ for every i , and
- (b) $S_\rho < \int_Q F$ for each of the three root lines.

Moreover, if (a) holds then $S_\alpha + S_\beta + S_{\alpha+\beta} = 2 \int_Q F$, so (b) states exactly that the three line shares satisfy the strict triangle inequality.

Proof. Every ray-spanned subspace V meets $N_{f,\mathbb{C}}$ in 0, a root line, or $N_{f,\mathbb{C}}$: a single lift $f_i + t_i$ or $-f_i$ contributes no fibre direction to a span, while the pair contributes exactly $\mathbb{C}t_i$.

Single lifts never help. If V contains single lifts, its ray set splits off theirs, and $\mu(\mathcal{F}_V)$ is a weighted mean of the slope of the remaining part and of base degrees $(n-1)! \int_Q F / (2 + \ell_{t_j})$, each strictly below $\mu(T_X)$ because $2 + \ell_{t_j} > 1$ off a set of measure zero. In particular a subspace spanned by single lifts alone (the case $V \cap N_{f,\mathbb{C}} = 0$) is subcritical. Thus, for testing stability, it suffices to consider configurations built from pairs and roots only.

Case $V \supseteq N_{f,\mathbb{C}}$, with pair set $A \subsetneq \{1, \dots, k\}$: Lemma 7.4 and (3) turn $\mu(\mathcal{F}_V) < \mu(T_X)$ into $\sum_{i \notin A} \varphi_i < 0$. As A varies these conditions are equivalent, termwise, to (a): taking A to omit a single index j shows $\varphi_j < 0$ is necessary, and conversely under (a) every sum $\sum_{i \notin A} \varphi_i$ of negative terms is negative.

Case $V \cap N_{f,\mathbb{C}} = \mathbb{C}\rho$, with pair set $A \subseteq \{i : t_i = \pm\rho\}$: the condition reads $\int_{E_\rho} F + \int_{E_{-\rho}} F - \sum_A \varphi_i < \int_Q F$, and the worst A gives (b).

The share identity. The six edge integrals sum to $\int_{\partial Q} F$, every twist lies on exactly one root line, and under (a) the corrections total $-\Phi$; so (3) gives $\sum_\rho S_\rho = (2 \int_Q F + \Phi) - \Phi = 2 \int_Q F$. $\square$

In $X(\alpha, \beta, -(\alpha+\beta))$, for instance, the maximal slope 506 against $\mu = 516$ is the N_f -configuration omitting one twist, and the gap 10 equals $-6\varphi_1$ (the three φ_i coincide by rotation symmetry). We now prove that the criterion is decisive for every zero-sum multiset.

Put

$$\gamma_1 = \alpha, \quad \gamma_2 = \beta, \quad \gamma_3 = -(\alpha + \beta), \quad x_j = \ell_{\gamma_j};$$

thus $x_1 + x_2 + x_3 = 0$ and $Q = \{|x_j| \leq 1\}$. If p_j, q_j are the multiplicities of $\gamma_j, -\gamma_j$, respectively, then the twist sum vanishes exactly when

$$p_1 - q_1 = p_2 - q_2 = p_3 - q_3 =: d.$$

Global negation of the roots is induced by the automorphism -1 of the hexagon and gives an isomorphic root-twist variety. Applying it if necessary, we may assume $d \geq 0$. Hence every

nonempty zero-sum multiset has the unique normal form

$$(5) \quad F = H^d \prod_{j=1}^3 (4 - x_j^2)^{q_j}, \quad H = \prod_{j=1}^3 (2 + x_j), \quad d, q_j \in \mathbb{Z}_{\geq 0}.$$

For $0 < a_i \leq 1$, write

$$K(a_1, a_2, a_3) = \{x_1 + x_2 + x_3 = 0 : |x_i| \leq a_i\}.$$

The normalization of Lebesgue measure on this plane will not matter in the next two sign lemmas.

Lemma 7.7 (positive-root box inequality). *For $d \geq 1$ and $j \in \{1, 2, 3\}$,*

$$\int_{K(a_1, a_2, a_3)} \frac{x_j H^d}{2 + x_j} dx < 0.$$

Proof. By relabeling take $j = 1$. Pair opposite polar rays of the centrally symmetric convex body K . Normalize a nontrivial pair as $tw = (at, bt, ct)$, $0 \leq t \leq 1$, where $a > 0$, $a + b + c = 0$, and $|a|, |b|, |c| \leq 1$. Apart from a positive angular factor, its contribution is

$$(6) \quad J = a \int_0^1 t^2 \{P(t) - P(-t)\} dt, \quad P(t) = (2 + at)^{d-1} (2 + bt)^d (2 + ct)^d.$$

If $bc \leq 0$, then, for $L = \log(P(t)/P(-t))$,

$$L' = 4 \left(d \sum_{\xi \in \{a, b, c\}} \frac{\xi}{4 - t^2 \xi^2} - \frac{a}{4 - t^2 a^2} \right),$$

and putting the first sum over a common denominator gives

$$\sum_{\xi \in \{a, b, c\}} \frac{\xi}{4 - t^2 \xi^2} = \frac{abc t^2 (12 - \frac{1}{2}(a^2 + b^2 + c^2)t^2)}{\prod_{\xi \in \{a, b, c\}} (4 - t^2 \xi^2)} \leq 0.$$

Thus $L' < 0$, and $P(t) < P(-t)$ for $t > 0$.

Suppose $bc > 0$. Write $b = -u$, $c = -v$, $a = u + v$, and $\omega = uv \leq a^2/4$. Then $(2 + bt)(2 + ct) = 4 - 2at + \omega t^2$, so $P(t) = (2 + at)^{d-1} (4 - 2at + \omega t^2)^d$ and $\partial_\omega P(t) = d t^2 H_\omega(t)^{d-1}$, where

$$H_\omega(t) = (2 + at)(4 - 2at + \omega t^2) = 8 - 2(a^2 - \omega)t^2 + a\omega t^3.$$

Regarding (6) as $J(a, \omega)$ and differentiating gives

$$\partial_\omega J = ad \int_0^1 t^4 \{H_\omega(t)^{d-1} - H_\omega(-t)^{d-1}\} dt \geq 0.$$

It is therefore enough to take $u = v = a/2$. With $q = at$, $J(a, a^2/4) = a^{-2} K_d(a)$, where

$$K_d(a) = \int_0^a q^2 \left[(2 + q)^{d-1} (2 - q/2)^{2d} - (2 - q)^{d-1} (2 + q/2)^{2d} \right] dq.$$

Using $z = (2 + q)/6$ in the first term and $z = (2 - q)/6$ in the second, $K_d(a)$ is a positive constant times the mass to the right of $1/3$ minus the mass to its left for

$$f(z) = (z - 1/3)^2 z^{d-1} (1 - z)^{2d}.$$

Put $r = z/(1 - z)$. The involution $r \mapsto r' = 1/(4r)$ maps the right interval into the left one. Including $|dr'/dr| = 1/(4r^2)$, the ratio of the transported left density to the right density is

$$\mathcal{Q}(r) = \frac{4^{2d+2}r^{d+1}(1+r)^{3d+3}}{(4r+1)^{3d+3}}.$$

Here $\mathcal{Q}(1/2) = 1$ and

$$(\log \mathcal{Q})' = \frac{(d+1)(2r-1)^2}{r(1+r)(4r+1)} > 0 \quad (r > 1/2).$$

Thus the left mass is strictly larger, so $K_d(a) < 0$. Every paired ray with $a \neq 0$ contributes negatively; the directions with $a = 0$ contribute zero and have angular measure zero. This proves the lemma. $\square$

Lemma 7.8 (negative-root box inequality). *For $d \geq 1$ and $j \in \{1, 2, 3\}$,*

$$\int_{K(a_1, a_2, a_3)} x_j H^d(2 + x_j) dx > 0.$$

Proof. Take $j = 1$, put $b = a_2$, $c = a_3$, and pair the slices $x_1 = s$ and $x_1 = -s$. For $0 < s < \min(a_1, b + c)$, set

$$J_s = [-b, b] \cap [s - c, s + c], \quad h(s) = \int_{J_s} (2 + x)^d (2 + s - x)^d dx.$$

The paired contribution is positive precisely when

$$(7) \quad \frac{h(s)}{h(-s)} < \left(\frac{2+s}{2-s} \right)^{d+1}.$$

For $s \geq b + c$ both slices have measure zero and contribute nothing. Reflection in the integral for $h(-s)$ puts both integrals on J_s . On this interval define

$$B = (2 - x)(2 - s + x) = C + sx - x^2, \quad C = 2(2 - s), \quad I_m = \int_{J_s} B^m dx.$$

Since $(2 + x)(2 + s - x) = B + 4s$,

$$(8) \quad \frac{h(s)}{h(-s)} = \sum_{k=0}^d \binom{d}{k} (4s)^k \frac{I_{d-k}}{I_d}.$$

We claim that, for $m \geq 1$,

$$(9) \quad \frac{I_{m-1}}{I_m} \leq \frac{m+1}{mC}.$$

The vertex of B is $s/2$. If J_s contains it, write $B = q - y^2$, where $q = (2 - s/2)^2 = C + s^2/4$. Integrating $(yB^m)'$ and using that the endpoints straddle $y = 0$ gives

$$(2m+1)I_m \geq 2mqI_{m-1}.$$

This implies (9), since $2q(m+1) - C(2m+1) = C + (m+1)s^2/2 > 0$.

If J_s misses the vertex, reflection $x \mapsto s - x$ interchanges b and c , so assume J_s lies to its left. Then $J_s = [L, b] \subset [-b, b]$, and B is increasing there. The quotient I_{m-1}/I_m is the B^m -weighted mean of $1/B$; deleting the low- B prefix $[-b, L]$ cannot increase it. It remains to

prove (9) on $[-b, b]$. Put $P = C - x^2$ and expand $T = (m+1)I_m - mCI_{m-1}$ in even powers of s . For $n = m - 2r \geq 1$, the coefficient of s^{2r} , apart from the positive factor $\binom{m}{2r}$, is

$$\begin{aligned} & \int_{-b}^b x^{2r} P^{n-1} \{(2r+1)C - (m+1)x^2\} dx \\ &= [x^{2r+1} P^n]_{-b}^b + n \int_{-b}^b x^{2r+2} P^{n-1} dx > 0. \end{aligned}$$

If $n = 0$, the final coefficient is plainly positive. Thus $T > 0$, and (9) follows.

Iteration gives

$$\frac{I_{d-k}}{I_d} \leq C^{-k} \frac{d+1}{d-k+1}.$$

Substitution in (8) yields

$$\frac{h(s)}{h(-s)} \leq \sum_{k=0}^d \binom{d+1}{k} \left(\frac{4s}{C}\right)^k < \left(1 + \frac{4s}{C}\right)^{d+1} = \left(\frac{2+s}{2-s}\right)^{d+1}.$$

The omitted $k = d+1$ term makes the inequality strict. This proves (7), hence the lemma. $\square$

The line-share inequality needs a sharper derivative form of the same convolution comparison.

Lemma 7.9 (box-share inequality). *For $a, b, c \in (0, 1]$ and an integer $d \geq 0$, put*

$$K_{b,c}(x) = \int_{[-b,b] \cap [-x-c, -x+c]} ((2+u)(2-x-u))^d du$$

and

$$\mathcal{D}_d(a, b, c) = - \int_{-a}^a x(2+x)^d K'_{b,c}(x) dx.$$

The function $K_{b,c}$ is Lipschitz (indeed, piecewise polynomial), and $K'_{b,c}$ denotes its almost-everywhere derivative. Then $\mathcal{D}_d(a, b, c) > 0$ for $d \geq 1$. For $d = 0$ it is nonnegative, and $\mathcal{D}_0(1, 1, 1) = 1$.

Proof. Assume first $d \geq 1$. Set $J(s) = K_{b,c}(-s)$ and pair s with $-s$:

$$(10) \quad \mathcal{D}_d(a, b, c) = \int_0^a sG(s) ds, \quad G(s) = (2+s)^d J'(-s) - (2-s)^d J'(s).$$

The integrand vanishes for $s \geq b+c$, so take $0 < s < b+c$; interchange b, c so that $b \leq c$. On

$$I = [-b, b] \cap [s-c, s+c] = [L, U]$$

write

$$A(x) = (2+x)(2+s-x), \quad B(x) = (2-x)(2-s+x), \quad m = d-1,$$

and $M_A = \int_I A^m$, $M_B = \int_I B^m$. If $s < c-b$, then $I = [-b, b]$ and

$$J'(s) = d \int_I (2+x) A^m, \quad J'(-s) = d \int_I (2-x) B^m.$$

If $c-b < s < b+c$, then $L = s-c$, $U = b$, and

$$J'(s) = -A(L)^d + d \int_I (2+x) A^m, \quad J'(-s) = B(L)^d + d \int_I (2-x) B^m.$$

Since $A' = B' = s - 2x$, integration by parts yields

$$d \int_I (2-x) B^m = \frac{d(4-s)}{2} M_B + \frac{[B^d]_L^U}{2}, \quad d \int_I (2+x) A^m = \frac{d(4+s)}{2} M_A - \frac{[A^d]_L^U}{2}.$$

The main term of $G(s)$ is therefore

$$(11) \quad \frac{d}{2} ((2+s)^d (4-s) M_B - (2-s)^d (4+s) M_A).$$

The remaining boundary term in the central case is

$$\frac{1}{2} \{ (2+s)^d (B(U)^d - B(L)^d) + (2-s)^d (A(U)^d - A(L)^d) \} > 0,$$

because $U = b, L = -b$ and both differences of the bases equal $2bs$. In the clipped case it is

$$\frac{1}{2} \{ (2+s)^d (B(L)^d + B(U)^d) + (2-s)^d (A(L)^d + A(U)^d) \} > 0.$$

It remains to prove, with $r = (2+s)/(2-s)$,

$$(12) \quad \frac{M_A}{M_B} < r^d \frac{4-s}{4+s}.$$

First suppose that I contains $h = s/2$. With $y = x - h$, $B = q - y^2$, $q = (2 - s/2)^2$, and $A = B + 4s$. For $N_j = \int_I B^j$, integration of $(yB^j)'$ gives

$$(2j+1)N_j - 2jqN_{j-1} = [yB^j]_{\partial I} \geq 0,$$

where the endpoints straddle $y = 0$. Iterating and expanding $(B + 4s)^m$ gives

$$\frac{M_A}{M_B} \leq S_m(z) := \sum_{k=0}^m \binom{m + \frac{1}{2}}{k} z^k, \quad z = \frac{4s}{q}.$$

This is the degree- m Taylor polynomial at zero of $(1+z)^{m+1/2}$, whose Lagrange remainder is positive. Hence, with $w = (4+s)/(4-s)$ and $1+z = w^2$,

$$\frac{M_A}{M_B} < w^{2m+1} < \frac{r^{m+1}}{w}.$$

The last inequality follows from $w^2 < r$; clearing denominators leaves the positive difference $2s^3$. Since $d = m + 1$, this is (12).

Suppose next that the central interval $[-b, b]$ misses h , so $b < h$. Pair x and $-x$ for $0 \leq x \leq b$. Put

$$D_0 = 4 - 2s - x^2, \quad z_0 = sx, \quad p = \frac{D_0 - z_0}{D_0 + z_0}, \quad P = 1 - p,$$

and

$$\theta = \frac{s-x}{2+s}, \quad \kappa = \frac{2-x}{2+s}.$$

The exact identities

$$\frac{A(x)}{rB(x)} = 1 - \theta P, \quad \frac{A(-x)}{rB(x)} = 1 - \kappa P$$

hold. Here $0 < p \leq 1$, $P = 1 - p$, and $0 \leq \theta, \kappa \leq 1$. Convexity gives the chord bound

$$(1 - \lambda P)^m \leq 1 - \lambda + \lambda p^m \quad (0 \leq \lambda \leq 1),$$

with $m = 0$ immediate. Applying this to θ and κ , using $\theta + \kappa = 1 - 2x/(2 + s) \leq 1$, and dividing by $1 + p^m$ gives

$$\frac{A(x)^m + A(-x)^m}{B(x)^m + B(-x)^m} \leq r^m \left(1 + \frac{2x}{2 + s}\right) \leq r^m \left(1 + \frac{s}{2 + s}\right) < r^{m+1} \frac{4 - s}{4 + s};$$

the final gap is $(s + 6)s^2/((2 - s)(2 + s)(4 + s)) > 0$. Integration proves (12). Finally, if a clipped interval misses h , then it lies to the left of h and is a suffix of $[-b, b]$. There B increases while $(A/B)^m = (1 + 4s/B)^m$ decreases. Deleting the low- B prefix decreases the B^m -weighted average M_A/M_B , reducing to the central case. Thus (12) always holds. The main term (11) and the positive boundary term give $G(s) > 0$, and (10) proves the assertion for $d \geq 1$.

For $d = 0$, $K_{b,c}$ is the convolution of two centered interval indicators, hence is even and nonincreasing on $[0, \infty)$. Thus $-xK'_{b,c}(x) \geq 0$. If $a = b = c = 1$, then $K_{1,1}(x) = 2 - |x|$ on $[-1, 1]$, and $\mathcal{D}_0(1, 1, 1) = 1$. $\square$

Proof of Theorem 7.2. We apply the three box inequalities to (5). For $q = 0$ let $\mu_q = \delta_1$, while for $q \geq 1$ put

$$(13) \quad d\mu_q(a) = 3^q d\delta_1(a) + 2qa(4 - a^2)^{q-1} da \quad (0 < a < 1).$$

Then, exactly on $|u| \leq 1$,

$$(14) \quad (4 - u^2)^q = \int_{(0,1]} \mathbf{1}_{\{|u| \leq a\}} d\mu_q(a).$$

Thus every product of pair factors is a positive mixture of indicators of the centered boxes $K(a_1, a_2, a_3)$.

Assume first $d \geq 1$. A positive-root occurrence on line j has singleton integral $\int_Q Fx_j/(2 + x_j)$, which is negative by (14) and Lemma 7.7. A negative-root occurrence requires $q_j \geq 1$. Peeling off one factor $4 - x_j^2 = (2 + x_j)(2 - x_j)$ rewrites its singleton as

$$- \int_Q x_j H^d(2 + x_j)(4 - x_j^2)^{q_j-1} \prod_{i \neq j} (4 - x_i^2)^{q_i},$$

which is negative by Lemma 7.8. If $d = 0$, then F is centrally even, and pairing x with $-x$ gives, for either occurring sign,

$$\varphi_{\pm\gamma_j} = - \int_Q F \frac{x_j^2}{4 - x_j^2} < 0.$$

This proves condition (a) of Proposition 7.6.

It remains to prove the line-share inequalities. Fix the first line. On $[-1, 1]$ set

$$w_j(u) = (2 + u)^d(4 - u^2)^{q_j}, \quad g_j(u) = w_j(u)\mathbf{1}_{\{|u| \leq 1\}},$$

and extend only g_j by zero to $\mathbb{R}$. Put

$$K(x) = \int_{\mathbb{R}} g_2(y)g_3(-x - y) dy, \quad I = \int_{-1}^1 w_1(x)K(x) dx = \int_Q F.$$

The two edge integrals on this line sum to

$$B_1 = w_1(-1)K(-1) + w_1(1)K(1) = \int_{E_{\gamma_1}} F + \int_{E_{-\gamma_1}} F.$$

Since the multiplicities on the line are $d + q_1$ copies of γ_1 and q_1 copies of $-\gamma_1$,

$$\sum_{t_i=\pm\gamma_1} \varphi_i = \int_{-1}^1 x w_1 K \left(\frac{d+q_1}{2+x} - \frac{q_1}{2-x} \right) dx = \int_{-1}^1 x w_1' K dx.$$

By condition (a), the share $S_1 := S_{\gamma_1}$ is $B_1 - \sum_{t_i=\pm\gamma_1} \varphi_i$. Since K is absolutely continuous, integration by parts on $[-1, 1]$ gives

$$(*) \quad B_1 = I + \int_{-1}^1 x(w_1' K + w_1 K') dx, \quad I - S_1 = - \int_{-1}^1 x w_1 K' dx.$$

The layer kernels are uniformly compactly supported and Lipschitz, so substitution of (14) in (*) and Fubini's theorem, applied also to their almost-everywhere derivatives, give

$$I - S_1 = \int_{(0,1]^3} \mathcal{D}_d(a, b, c) d\mu_{q_1}(a) d\mu_{q_2}(b) d\mu_{q_3}(c).$$

For $d \geq 1$ this is strictly positive by Lemma 7.9. For $d = 0$ every layer is nonnegative, while the atom $(a, b, c) = (1, 1, 1)$ has positive mass and contributes 1. Thus $S_1 < I$ in all cases. The same argument on the other two root lines proves condition (b), so Proposition 7.6 gives stability.

Finally, for $k = n - 2 \geq 2$, use $k/2$ antipodal pairs if k is even, and one rotation triple together with $(k-3)/2$ antipodal pairs if k is odd. These zero-sum multisets give the asserted examples in every dimension. $\square$

Lemma 7.10 (extreme-exponent ordering). *Assume $d \geq 1$ in (5), and put*

$$M_i = \int_Q x_i F(x) dx \quad (i = 1, 2, 3).$$

If $q_i = \max_{\nu} q_{\nu}$, $q_j = \min_{\nu} q_{\nu}$, and $q_i > q_j$, then $M_i < M_j$. In particular, the hexagon moment can vanish only if $q_1 = q_2 = q_3$.

Proof. Write

$$E = 8 - \sum_{\nu=1}^3 x_{\nu}^2, \quad e = x_1 x_2 x_3, \quad G = \prod_{\nu=1}^3 (4 - x_{\nu}^2)^{q_{\nu}},$$

so that $H = E + e$ and $H(-x) = E - e$. The divided difference

$$R_d = \sum_{m=0}^{\lfloor (d-1)/2 \rfloor} \binom{d}{2m+1} E^{d-2m-1} e^{2m} = \frac{(E+e)^d - (E-e)^d}{2e}$$

where the quotient at $e = 0$ is interpreted as dE^{d-1} , is continuous and strictly positive on Q . It is invariant under coordinate permutations and central inversion. Since G is centrally even, central symmetry gives

$$(15) \quad M_i = \frac{1}{2} \int_Q x_i (H(x)^d - H(-x)^d) G(x) dx = \int_Q x_i e R_d G dx.$$

Up to a null set, Q is the disjoint union of the twelve images, under coordinate permutations and central inversion, of

$$\mathcal{T} = \{(a, b) : 0 < a < b, a + b < 1\}, \quad (x_1, x_2, x_3) = (a, b, -c), \quad c = a + b.$$

These maps have Jacobian of absolute value one in lattice coordinates. Put

$$A = 4 - a^2 > B = 4 - b^2 > C = 4 - c^2 > 0,$$

and attach the scores $s_A = -a$, $s_B = -b$, $s_C = c$ to the corresponding even factors $U_A = A$, $U_B = B$, $U_C = C$. Define

$$V_i = \sum_{\sigma: \{1,2,3\} \xrightarrow{\sim} \{A,B,C\}} s_{\sigma(i)} \prod_{\nu=1}^3 U_{\sigma(\nu)}^{q_\nu}.$$

At $(a, b, -c)$ one has $e = -abc$, and assigning the magnitude labelled L to coordinate i gives $x_i e = abc s_L$. The negative sector gives the same product. Hence (15) becomes

$$(16) \quad M_i = 2 \int_{\mathcal{T}} abc R_d(a, b, -c) V_i(a, b) da db.$$

Let k be the remaining index and write

$$q_i = q + r, \quad q_j = q, \quad q_k = q + h, \quad r > 0, \quad 0 \leq h \leq r.$$

Pairing in $V_i - V_j$ each assignment with the one that interchanges the labels at i and j , and grouping by the label at k , gives

$$\begin{aligned} V_i - V_j = (ABC)^q & [(b-a)(A^r - B^r)C^h \\ & - (2a+b)(A^r - C^r)B^h \\ & - (a+2b)(B^r - C^r)A^h]. \end{aligned}$$

The absolute value of the second term alone is strictly larger than the first, since

$$2a+b > b-a, \quad A^r - C^r > A^r - B^r, \quad B^h \geq C^h,$$

and the third term is also strictly negative. Thus $V_i < V_j$ throughout $\mathcal{T}$. Equation (16) and $R_d > 0$ imply $M_i < M_j$. $\square$

Proof of Theorem 7.3. Use the normal form (5), after global negation if necessary; this preserves moment vanishing and both symmetry alternatives. If $d = 0$, the multiplicities of every root and its negative agree. Thus the multiset is negation-invariant, F is centrally even, and its hexagon moment vanishes.

Suppose $d \geq 1$. If $q_1 = q_2 = q_3$, then the multiplicities $d+q_j$ of the γ_j and the multiplicities q_j of the $-\gamma_j$ are separately constant in j . Hence the multiset and F are invariant under τ . The three numbers $M_j = \int_Q x_j F$ are equal; since $x_1 + x_2 + x_3 = 0$, they vanish. Conversely, if the hexagon moment vanishes, then every M_j vanishes, and Lemma 7.10 forces $q_1 = q_2 = q_3$. For $d \geq 1$ the multiset cannot be negation-invariant, because the multiplicities of γ_j and $-\gamma_j$ differ by d .

The moment therefore vanishes exactly in the two stated symmetry cases. Lemma 7.4(3) identifies moment vanishing with the Kähler–Einstein condition. $\square$

The general result contains two of the following three explicit families; the third shows that zero sum is not necessary.

Theorem 7.11. *For every $r \geq 1$, $m \geq 1$, and $s \geq 0$, the root-twist varieties*

$$X((\alpha, -\alpha)^{\times r}) \quad (n = 2r + 2) \quad \text{and} \quad X((\alpha, \beta, -(\alpha + \beta))^{\times m}) \quad (n = 3m + 2)$$

have $(-K)$ -stable tangent bundle and are Kähler–Einstein. The varieties

$$Y_s = X((\alpha, -\alpha)^{\times s}, \alpha, \beta) \quad (n = 2s + 4)$$

also have stable tangent bundle but are not Kähler–Einstein. All three families are smooth toric Fanos of Picard rank $n+2$. In particular, stable Kähler–Einstein examples of rank $n+2$

exist in every even dimension $n \geq 4$ and every dimension $n \equiv 2 \pmod{3}$, $n \geq 5$, while stable non-Kähler-Einstein examples of that rank exist in every even dimension $n \geq 4$.

Proof. Smooth Fano is Lemma 7.4(1).

Kähler-Einstein. The pair and rotation-orbit multisets satisfy the two symmetry alternatives of Theorem 7.3.

Stability. The pair and triple twist multisets are nonzero and zero-sum, so their tangent bundles are stable by Theorem 7.2.

The unbalanced family. Put $x = \ell_\alpha$, $y = \ell_\beta$, $q(u) = 4 - u^2$, and $M_j = \int_0^1 u^j q(u)^s du$. For Y_s one has $F = q(x)^s(2+x)(2+y)$. For $0 \leq u \leq 1$, the slices of Q at $x = u$ and $x = -u$ are respectively $[-1, 1-u]$ and $[-1+u, 1]$ in the y -coordinate, and

$$\int_{-1}^{1-u} (2+y) dy = 4 - 3u + \frac{1}{2}u^2, \quad \int_{-1+u}^1 (2+y) dy = 4 - u - \frac{1}{2}u^2.$$

Consequently, with $I_0 = \int_Q F$,

$$I_0 = 16M_0 - 8M_1 - 2M_2 + M_3, \quad \varphi_\alpha = \varphi_\beta = -2M_2 + M_3 = -\int_0^1 u^2(2-u)q(u)^s du < 0.$$

If $s \geq 1$, the remaining singleton integral is

$$\varphi_{-\alpha} = -\int_0^1 u^2 q(u)^{s-1} (24 - 12u - 2u^2 + u^3) du < 0,$$

since the last factor is at least 11 on $[0, 1]$. Thus Proposition 7.6(a) holds (and for $s = 0$ there is no $-\alpha$ twist).

Let $B_\rho = \int_{E_\rho} F + \int_{E_{-\rho}} F$. Direct edge integration gives

$$B_\beta = 8M_0 - 2M_1, \quad B_{\alpha+\beta} = 8M_0 + 2M_1 - 2M_2.$$

There is one β twist and no $\pm(\alpha + \beta)$ twist, hence

$$S_\beta = 8M_0 - 2M_1 + 2M_2 - M_3, \quad S_{\alpha+\beta} = 8M_0 + 2M_1 - 2M_2.$$

It follows that

$$I_0 - S_\beta = 2 \int_0^1 q(u)^s (1-u)(4+u-u^2) du > 0.$$

The functions $q(u)^s$ and $h(u) = 8 - 10u + u^3$ are nonincreasing on $[0, 1]$. Chebyshev's integral inequality therefore gives

$$I_0 - S_{\alpha+\beta} = \int_0^1 q(u)^s h(u) du \geq M_0 \int_0^1 h(u) du = \frac{13}{4}M_0 > 0.$$

Finally the share identity yields

$$I_0 - S_\alpha = S_\beta + S_{\alpha+\beta} - I_0 = 2 \int_0^1 q(u)^s (4u + u^2 - u^3) du > 0.$$

Proposition 7.6(b) now proves stability.

On the other hand, central symmetry of Q gives

$$\int_Q xF = 2 \int_Q q(x)^s (x^2 + xy) = 2 \int_0^1 u^2(2-u)q(u)^s du > 0.$$

The hexagon moment is therefore nonzero, so Lemma 7.4(3) shows that Y_s is not Kähler-Einstein. $\square$

The first unbalanced member $Y_0 = X(\alpha, \beta)$ is the census entry F.4D.0061: it has $(-K)^4 = 278$, $\mu(T_X) = 139/2$, and maximal subsheaf slope $206/3$, with barycenter $(10, 10, -5, -5)/139$ in the construction coordinates. Explicitly, writing rays as row vectors, right multiplication by the unimodular matrix

$$\begin{pmatrix} 0 & 0 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

sends its ten rays onto the stored rays of F.4D.0061. Thus the failure of twist-sum necessity is already visible in dimension four, and Theorem 7.11 propagates it to every even dimension.

Remark 7.12. The margins are explicit: for the triple family every share sits uniformly $\frac{1}{3} \int_Q F$ below critical, while for the pair family the margin on the twisted line is $2J = (4^{r+1} - 3^{r+1})/(r+1) = o(\int_Q F)$, where $J = \int_0^1 u(4-u^2)^r du$: the one-line family is asymptotically critical on its own line, consistent with the near-critical maximal slopes these varieties exhibit in the census. (For the asymptotic claim, the numerator is $O(4^r/r)$, while slicing over $|\ell_\alpha| \leq r^{-1/2}$ gives $\int_Q F \gg 4^r/\sqrt{r}$.)

The normal form (5) and the positive decomposition (14) show why the mixed cases remain tractable despite their lack of symmetry: the relevant inequalities are preserved under positive averaging on centered-box layers; when $d = 0$, the full-box atom supplies the required strictness. Thus zero sum is sufficient for stability in all dimensions, while the unbalanced family shows that it is not necessary.

By Lemma 7.4(3), the Kähler–Einstein question is the moment problem $\int_Q xF(x) dx = 0$. Theorem 7.3 solves it under the zero-sum hypothesis; classifying vanishing moments for arbitrary root multisets remains open.

8. STRUCTURAL OBSERVATIONS AND OPEN QUESTIONS

8.1. Antipodal directions in destabilizers. Steffens proved that a Fano three-fold whose tangent bundle is not stable admits a destabilizing subsheaf that is the relative tangent sheaf of the contraction of an extremal face, and expected the statement to persist in every dimension [Ste96]. For every unstable case our computation forms the sum of all maximal-slope ray-spanned subspaces, which is the canonical first Harder–Narasimhan subspace. Its dimension is one in 5/7, 38/74, 375/675, 3865/6620 of the unstable cases in dimensions 3, 4, 5, 6. It contains an antipodal pair $\pm u$ of rays in 7/7, 68/74, 558/675, 4877/6620 of them. Such a pair is the Klyachko datum of a rank-one relative tangent candidate; determining whether it comes from an equivariant $\mathbb{P}^1$ -fibration requires checking compatibility with the fan. The no-pair cases (six in dimension four, 117 in dimension five, 1743 in dimension six) include HN subspaces spanned by a single ray (1, 50, 986 in dimensions 4, 5, 6), candidates for relative tangent sheaves of birational contractions. Matching these subspaces to extremal faces of the Mori cone is needed to turn the numerical pattern into a theorem or a counterexample to the proposed higher-dimensional principle.

8.2. High Picard rank in the Fano census. Question 1.5 of [HNS22] asks whether, for fixed (n, ρ) , only finitely many smooth projective toric n -folds of Picard rank ρ have a polarization making T_X stable. The present census does not address that question: there are only finitely many smooth toric Fano n -folds for each n , so finiteness among anticanonical Fanos is immediate. What the tables of Section 4 do show is the shape of the stable locus at $-K$:

it thins rapidly with ρ , and its high- ρ boundary is occupied by a single structured example in dimension five (Section 7). The root-twist varieties have $\rho = n + 2$, and Theorem 7.2 gives stable examples at that rank in every dimension $n \geq 4$; whether they exhaust the high-rank stable locus is open.

8.3. Open questions.

- (1) Classify arbitrary root multisets, without the zero-sum hypothesis, whose hexagon moment vanishes. More generally, characterize the toric fibrations for which a twist turns all slope equalities of the fiber–base decomposition strict.
- (2) Determine the maximal Picard rank of a stable example in each dimension. Is it $n + 2$, attained by the root-twist family, for all $n \geq 5$? By Theorem 7.2, rank $n + 2$ is attained in every dimension $n \geq 4$; only maximality is open.
- (3) Extend the census from $-K$ to the full nef cone (the stability chambers of [HNS22], Ex. 5.2), where already in dimension three the chamber structure is not polyhedral.
- (4) Quantify the asymptotics of the stable, polystable and Kähler–Einstein fractions as $n \rightarrow \infty$.

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